GAT Quantitative Comparisons & Fast Geometry

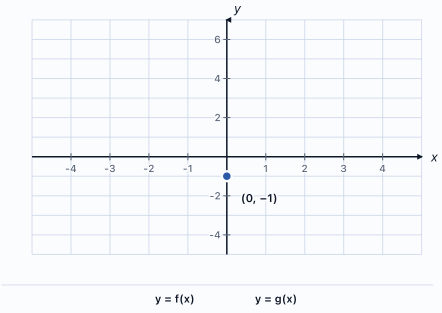
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

GAT Quantitative Comparisons & Fast Geometry Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 10x + 38$ in the domain of GAT Quantitative Comparisons & Fast Geometry. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(7)$.



A 35
B 17
C 30
D 28

2 MCQ

A mathematical model in GAT Quantitative Comparisons & Fast Geometry is defined by the cubic function $y_2(x) = 4x^3 - 5x^2 + 15x - 10$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123
B 78
C 41
D 93

3 MCQ

In GAT Quantitative Comparisons & Fast Geometry, two displacement vectors are given by $\mathbf{u}_3 = \langle 5, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 78
B -11
C 89
D 190

4 MCQ

The linear growth profile shown in the diagram for GAT Quantitative Comparisons & Fast Geometry follows an arithmetic progression with first term $a_1 = 6$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 90

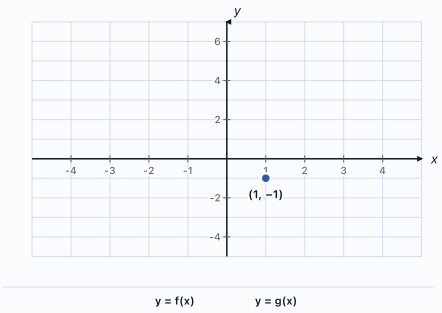
B 720

C 810

D 1350

5 MCQ

In GAT Quantitative Comparisons & Fast Geometry, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 20x)\,\mathrm{d}x$.



A 112

B 108

C 88

D 45

6 MCQ

In GAT Quantitative Comparisons & Fast Geometry, consider the quadratic function $f_6(x) = x^2 - 12x + 121$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(13)$.

A 225

B 134

C 212

D 219

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 6x^2 + 33x - 17$ studied in GAT Quantitative Comparisons & Fast Geometry. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 705

B 801

C 672

D 117

8 MCQ

In GAT Quantitative Comparisons & Fast Geometry, two displacement vectors are given by $\mathbf{u}_8 = \langle 6, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 13

B 98

C 85

D 168

9 MCQ

Compare the two quantities when $x = \frac{1}{3}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

10 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{4^2 + 7^2}$ | Quantity B: $4 + 7$

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

11 MCQ

Compare the two quantities: Quantity A: 12% of 35 | Quantity B: 35% of 12

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

12 MCQ

Given $x^2 = 16$ and $y = 3$, compare: Quantity A: x | Quantity B: y

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

13 MCQ

Compare the two quantities when $x = \frac{1}{4}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

14 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{5^2 + 8^2}$ | Quantity B: $5 + 8$

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

15 MCQ

Compare the two quantities: Quantity A: 14% of 38 | Quantity B: 38% of 14

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

16 MCQ

Given $x^2 = 25$ and $y = 4$, compare: Quantity A: x | Quantity B: y

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

17 MCQ

Compare the two quantities when $x = \frac{1}{5}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

18 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{6^2 + 9^2}$ | Quantity B: $6 + 9$

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

19 MCQ

Compare the two quantities: Quantity A: 16% of 41 | Quantity B: 41% of 16

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

20 MCQ

Given $x^2 = 36$ and $y = 5$, compare: Quantity A: x | Quantity B: y

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

21 MCQ

Compare the two quantities when $x = \frac{1}{6}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

22 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{7^2 + 10^2}$ | Quantity B: $7 + 10$

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

23 MCQ

Compare the two quantities: Quantity A: 18% of 44 | Quantity B: 44% of 18

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

24 MCQ

Given $x^2 = 49$ and $y = 6$, compare: Quantity A: x | Quantity B: y

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given

25 MCQ

Compare the two quantities when $x = \frac{1}{7}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

A Quantity A is greater

B Quantity B is greater

C The two quantities are equal

D The relationship cannot be determined from the information given